NativeScript: Creating a Mobile App Using JavaScript

This article explores NativeScript, an open source framework for building truly native mobile apps.

There are a number of options available in the market today for cross-platform mobile app development. In this article we explore NativeScript, which is an easy and user-friendly mobile app development platform. You can learn it very quickly if you know HTML, CSS and JavaScript. NativeScript mainly focuses on the front-end and uses Angular, Modern JavaScript and TypeScript.

Architecture

The NativeScript hybrid mobile app development architecture is divided into multiple layers. The mobile world is divided into two main platforms—Android and iOS. Each of these platforms has its own JavaScript machine. For iOS, it's JavaScript Core; this is a JavaScript VM that runs JavaScript code for iOS. Likewise, for Android, we have the V8, an environment that runs JavaScript code for Android.

The NativeScript runtime provides the native capability for both the platforms. By using NativeScript modules or packages, we can provide the various capabilities that will be available to the JavaScript app. Beyond that, we can build the app's code using Angular JS. We use JavaScript wrappers for native app development. The actual rendering of the UI is done using native UI elements. That's why it behaves similar to or as fast as vanilla Android or iOS applications.

Installation and a demo

NativeScript is built on Node.js; so if you don't have the latter, you need to install it. Installation is very simple. Go to the official website of Node.js, and get the installation file. Double click on it and install it. The link for the official website is *https://nodejs.org/*.

To check whether Node.js is already installed or not, you can use the following command:

```
node -v
```

If you get an error message like 'No command found' or 'Command is not recognised', it means you have not installed Node.js. After installing it, the next step is to install NativeScript, for which you can use the following command:

```
npm install -g nativescript
```

To verify whether NativeScript has been successfully installed, we can run the following command:

```
tns
```

Here, 'tns' stands for Telerik NativeScript since Telerik has developed this framework. If you see a lot of options as an output, it means you have successfully installed NativeScript.

After installing NativeScript, the next step is to install the requirements for iOS and Android. To build native iOS and Android apps, set up the development environment. Here, *tns* provides a quick start script which will help us to install everything.

Setting up the development environment in Windows: Press the *Start* button, type *cmd*, and then right-click and run as an administrator, going on to click the *Enter* button. Next, run the following script at the CLI prompt. Make sure that you run it as an administrator; else, it will fail or give some error.